

Use Case Diagram and Description

Vehicle Rental Management System (VRMS)

Definition

- A diagram that shows a set of use cases and actors and their relationships
- Use case – system functionality from the user's perspective

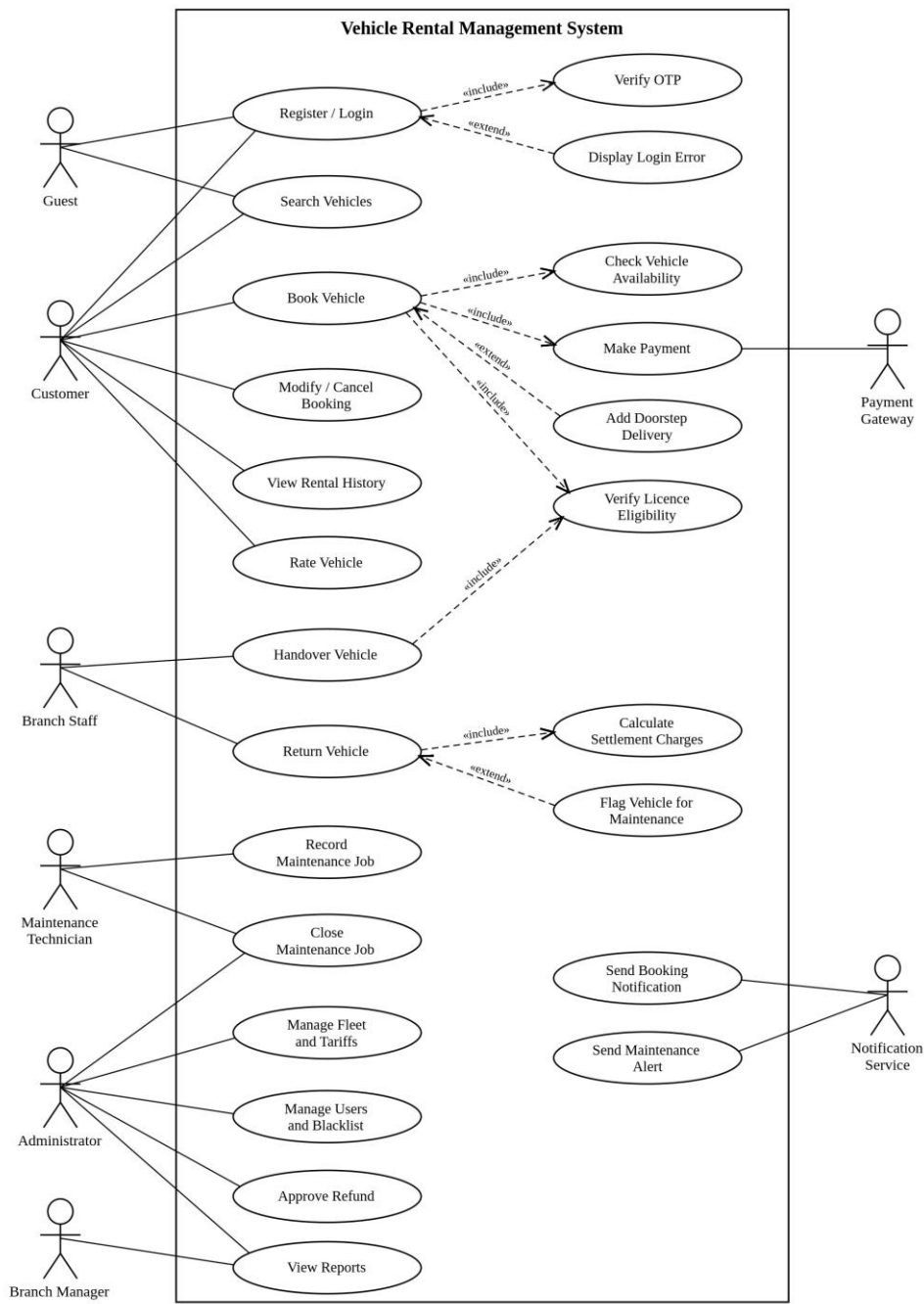


Figure 1. Use case diagram for the Vehicle Rental Management System.

Use Case Description

Use Case Name	<i>[Name of the use case]</i>
Actors	<i>[An actor is a person or other entity external to the system being specified who interacts with the system and performs use cases to accomplish tasks]</i>
Preconditions	<i>[Activities that must take place, or any conditions that must be true, before the use case can be started]</i>
Normal Flow – Description	<i>[User actions and system responses that will take place during execution of the use case under normal, expected conditions.]</i>
Normal Flow – Postconditions	<i>[State of the system at the conclusion of the use case execution with a normal flow (nominal)]</i>
Alternative flows and exceptions	<i>[Major alternative flows or exceptions that may occur in the flow of event]</i>
Non functional requirements	<i>[All non-functional requirement: e.g., dependability (safety, reliability, etc.), performance, ergonomic]</i>

Register / Login
Brief description: The actor creates an account or authenticates to access the system.
Actors: Guest, Customer
Preconditions: The actor has a mobile number capable of receiving an SMS
Basic flow of events: 1. The actor chooses to register or log in. 2. For registration, the actor supplies name, e-mail, mobile number, date of birth and a password. 3. The system verifies the mobile number by one-time password (include: Verify OTP). 4. For login, the actor supplies e-mail or mobile number and password. 5. The system authenticates the actor and opens the dashboard for that role.
Extensions: 3a. The one-time password entered is incorrect or has expired. • The system informs the actor and allows the actor to request a new code. 5a. The credentials are incorrect (extend: Display Login Error). • The system displays a generic error that does not reveal which credential was wrong.
Post-conditions: The actor holds an authenticated session appropriate to their role
Special requirements: Actor is connected to the Internet and uses a supported web browser or the mobile application

Search Vehicles
Brief description: The actor searches the fleet for a vehicle available at a chosen branch and period.
Actors: Customer, Guest
Preconditions: None (available to unauthenticated visitors for browsing)
Basic flow of events: 1. The actor enters a pickup branch, pickup date-time and return date-time. 2. The actor applies optional filters for type, category, price and rating. 3. The system computes which vehicles are genuinely available for the whole period. 4. The system displays the matching vehicles with an estimated price for each. 5. The actor selects a vehicle to view its full detail.

Extensions:
3a. No vehicle matches the criteria. The system states this and offers the nearest alternatives by branch or date.
Post-conditions: A list of available vehicles matching the criteria is displayed to the actor
Special requirements: None

Book Vehicle
Brief description: The actor reserves a car or two-wheeler for a specific pickup branch and rental period.
Actors: Customer
Preconditions: The actor is logged in and has a verified driving licence
Basic flow of events: - The actor selects a vehicle, pickup branch, pickup date-time and return date-time. - The system verifies the actor's eligibility (include: Verify Licence Eligibility). - The system checks that the vehicle is free for the requested period (include: Check Vehicle Availability). - The actor optionally selects add-ons, including doorstep delivery (extend: Add Doorstep Delivery). - The system places a 15-minute hold on the vehicle and displays an itemised quotation with the security deposit. - The actor selects a payment method and confirms (include: Make Payment). - The system stores the booking as Confirmed, assigns a booking reference and pickup code, and notifies the actor and the branch.
Extensions: 2a. The actor's licence is not verified, expired, or does not cover the vehicle category. - The system informs the actor and does not allow the booking to proceed. 3a. The vehicle is not available for the requested period. - The system returns the actor to the search results. 5a. The hold expires before payment is completed. - The system releases the vehicle and informs the actor. 6a. The payment authorisation is declined. - The system displays the reason and lets the actor retry or choose another method.

Post-conditions: A new booking exists with status Confirmed, the vehicle is unavailable for the booked period, and an invoice is generated

Special requirements: Actor is connected to the Internet and uses a supported web browser or the mobile application

Modify / Cancel Booking

Brief description: The actor changes the details of, or withdraws from, an existing booking.

Actors: Customer

Preconditions: The actor has a Confirmed booking that has not yet started

Basic flow of events:

1. The actor opens an existing Confirmed booking.
2. The actor chooses to modify the period, branch or add-ons, or to cancel the booking.
3. For a modification, the system checks that the vehicle is free for the new period and recalculates the price.
4. For a cancellation, the system displays the refund due under the cancellation policy.
5. The actor confirms the change.
6. The system updates or cancels the booking and notifies the actor and the branch.

Extensions:

- 2a. The request is made less than 12 hours before the pickup time.
 - The system refuses the modification and states the cut-off.
- 3a. The vehicle is not free for the new period.
 - The system offers an available vehicle of the same category at the same tariff.

Post-conditions: The booking reflects the new details, or its status is set to Cancelled and a refund has been initiated

Special requirements: None

View Rental History

Brief description: The actor reviews their past, active and upcoming rentals and downloads related documents.

Actors: Customer

Preconditions: The actor is logged in and has placed at least one booking
Basic flow of events:
1. The actor opens the rental history. 2. The system displays all of the actor's bookings, most recent first. 3. The actor filters or searches the list, or selects a specific booking. 4. The system displays the full detail, with links to the invoice and both condition reports. 5. The actor downloads a document, or chooses to rebook the same vehicle.
Extensions: 5a. The actor chooses to rebook. • The system opens a new booking pre-filled with that vehicle, subject to current availability and tariff.
Post-conditions: The requested document is downloaded, or a new booking is started
Special requirements: None

Rate Vehicle
Brief description: The actor rates the vehicle and the service after a completed rental.
Actors: Customer
Preconditions: The actor has a booking with status Completed, within the last 14 days
Basic flow of events: 1. The system invites the actor to rate the completed booking. 2. The actor rates the vehicle and the service separately on a scale of 1 to 5 and adds an optional comment. 3. The system stores the review and recalculates the vehicle's average rating.
Extensions: 2a. The actor has already reviewed this booking. • The system permits the actor to edit the existing review only within 24 hours of submission.
Post-conditions: The review is stored and the vehicle's average rating is updated
Special requirements: None

Handover Vehicle
Brief description: The actor records the physical release of a booked vehicle to the customer.

Actors: Branch Staff
Preconditions: The booking has status Confirmed and the pickup time has arrived
Basic flow of events: 1. The actor retrieves the booking by reference, pickup code or customer mobile number. 2. The system re-verifies the customer's licence eligibility (include: Verify Licence Eligibility). 3. The actor records the odometer reading, fuel level, accessories issued and photographs of the vehicle. 4. The customer signs the condition report on the branch device. 5. The system sets the booking to Active and the vehicle to On Rent, and e-mails the signed report to the customer.
Extensions: 2a. The licence has expired, or a statutory vehicle document is invalid for the rental period. • The system prevents handover and states which condition was not met.
Post-conditions: The booking status is Active and the vehicle status is On Rent
Special requirements: Actor uses a camera-equipped tablet with a touch screen for the signature

Return Vehicle
Brief description: The actor records the return of a vehicle and settles any additional charges.
Actors: Branch Staff
Preconditions: The booking has status Active
Basic flow of events: 1. The actor opens the return checklist for the booking. 2. The actor records the odometer reading, fuel level, accessories returned and photographs of the vehicle. 3. The system computes the itemised settlement (include: Calculate Settlement Charges). 4. The system presents the settlement to the customer for acknowledgement. 5. The system deducts the charges from the security deposit and refunds any balance. 6. The system sets the booking to Completed and the vehicle to Available.
Extensions: 2a. The return odometer reading is lower than the handover reading. • The system rejects the entry and requires correction.

3a. The staff member identifies new damage or a fault (extend: Flag Vehicle for Maintenance). The system creates a maintenance job and, if the vehicle is unfit for use, sets its status to Under Maintenance instead of Available.
Post-conditions: The booking status is Completed and the vehicle is released or sent for maintenance
Special requirements: Actor uses a camera-equipped tablet for the condition report

Record Maintenance Job
Brief description: The actor logs the work performed on a vehicle during a maintenance job.
Actors: Maintenance Technician
Preconditions: A maintenance job has been created and assigned to the actor
Basic flow of events: - The actor opens an assigned maintenance job. - The actor records the service date, odometer reading, work performed, parts used and labour cost. - The actor attaches the workshop invoice, if any. - The system stores the maintenance record against the vehicle's history.
Extensions: 2a. The job was created automatically by a preventive-service trigger. - The system pre-fills the job type as Preventive and the due distance or date.
Post-conditions: The maintenance record is stored and linked to the vehicle's history
Special requirements: None

Close Maintenance Job
Brief description: The actor completes a maintenance job and returns the vehicle to service.
Actors: Maintenance Technician, Administrator
Preconditions: The maintenance job has status In Progress and the recorded work is complete

Basic flow of events: 1. The actor reviews the recorded work, parts and cost for the job. 2. The actor closes the job. 3. The system sets the vehicle status back to Available unless it remains flagged.
4. The system computes and stores the next preventive service due, in distance and date.
Extensions: 1a. The vehicle was flagged Unsafe. • Only an Administrator or authorised technician may clear the flag as part of closing the job.
Post-conditions: The job status is Completed and the vehicle's next service due is updated
Special requirements: None

Manage Fleet and Tariffs
Brief description: The actor maintains the vehicle master data, branches and tariffs.
Actors: Administrator
Preconditions: The actor is logged in with the Administrator role
Basic flow of events: 1. The actor adds, edits or retires a vehicle, recording its registration, category, specification and documents. 2. The actor defines or changes a tariff for a vehicle category, with an effective date. 3. The system stores the change and records it in the audit log.
Extensions: 1a. The registration or chassis number already exists. • The system rejects the entry. 2a. A tariff change is requested. • The system retains the previous tariff so that existing confirmed bookings and issued invoices are unaffected.
Post-conditions: The fleet or tariff data is updated and an audit entry is recorded
Special requirements: None

Manage Users and Blacklist
Brief description: The actor administers staff accounts and restricts blacklisted customers.
Actors: Administrator
Preconditions: The actor is logged in with the Administrator role
Basic flow of events: 1. The actor creates, edits, disables or re-enables a Branch Staff, Technician or Branch Manager account. 2. The actor blacklists a customer, recording the reason. 3. The system applies the change immediately and records it in the audit log.
Extensions: 2a. The blacklisted customer has existing Confirmed bookings. • The system lists them so the actor may decide whether to cancel each one.
Post-conditions: The user account or blacklist entry is updated and an audit entry is recorded
Special requirements: None

Approve Refund
Brief description: The actor authorises a refund that exceeds the automatic approval threshold.
Actors: Administrator
Preconditions: A refund request is pending approval
Basic flow of events: 1. The actor reviews the pending refund, its amount and reason. 2. The actor approves or rejects the refund. 3. On approval, the system issues the refund to the original payment instrument and records the approver.
Extensions: 2a. The actor rejects the refund. • The system records the reason and notifies the customer.
Post-conditions: The refund is issued or rejected, and the decision is recorded in the audit log
Special requirements: None

View Reports

Brief description: The actor reviews utilisation, revenue, maintenance-cost and customer reports.
Actors: Administrator, Branch Manager
Preconditions: The actor is logged in with the Administrator or Branch Manager role
Basic flow of events: 1. The actor selects a report type and a period. 2. The system computes and displays the report. 3. The actor optionally exports the report to a file.
Extensions: 1a. The actor is a Branch Manager. • The system restricts the report to that actor's own branch.
Post-conditions: The requested report is displayed or exported
Special requirements: None